LESSON PLAN

ENGINE HEALTH MONITORING (9)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	07
Total No. of Lesson Plans	7	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Swaging, plumbing and flaring process and tools used
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Swaging, plumbing and flaring process and tools used
5. Specific Objectives	9.1.2.1 Trainee will be able to explain the locking and swaging etc.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Demonstration of swaging, plumbing, and flaring tools; safety precautions; instructor supervision
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary purpose of engine health monitoring?
(b) List three common types of hydraulic tubing used in aircraft.
(c) Why is proper flaring critical for system integrity?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Swaging process definition and application • Types of swaging tools and dies • Locking mechanisms in swaged joints • Safety precautions during swaging operations	L & Ds	PPT, IB	15 min
(b)	• Flaring process types (single and double) • Flaring tools and die selection • Plumbing assembly procedures • Inspection criteria for flared joints	L & Ds	PPT, PDF	20 min
(c)	• Tool maintenance and storage • Common defects in swaging and flaring • Troubleshooting plumbing issues • Documentation and record keeping	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the difference between single and double flaring.
(b) Describe the role of locking in a swaged joint.
(c) Identify two common defects caused by improper tool usage.

12. Home Assignment and Follow Up	Prepare a brief note on the next lesson plan (LP 02) focusing on hydraulic system pressure testing procedures.
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	End of subject	Refer to the master syllabus reference list	-

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min